

ABHISHEK JHA

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SUMMARY

Experienced Data Analyst with 3+ years of enterprise experience (EY) in data engineering, automated workflows, and BI analytics. Currently pursuing an MSc in Mathematical Modelling to transition a strong foundation in large-scale data processing into advanced machine learning. Passionate about using computational mathematics to build robust, data-driven AI solutions.

EDUCATION

University of Luxembourg

Master of Science in Mathematical Modelling and Computational Sciences

Luxembourg

Sep 2025 – Present

- **Academic Focus:** Rigorous probability theory, measure theory, numerical linear algebra (matrix factorizations, pseudoinverses), and finite element methods.

Delhi Technological University

Bachelor of Engineering in Mathematics and Computing

Delhi, India

Aug 2018 – Aug 2022

WORK EXPERIENCE

Ernst & Young LLP, India

July 2022 – August 2025

Data Analyst (Technology Consultant)

- Led data analysis and automated grievance workflows for large-scale Government of India healthcare projects, improving operational efficiency by **12.3%** and removing the need for manual intervention.
- Developed **RESTful APIs** using FastAPI to automate data extraction, allowing real-time data syncing across multiple national healthcare platforms.
- Built interactive **Power BI** dashboards and Python reporting pipelines (Pandas, Plotly) to provide real-time analytics to government stakeholders.
- Cleaned and processed large datasets (up to **1M+ records**) using Pandas, handling missing data and outliers to ensure data quality for analysis.

INeuron, Remote

Nov 2020 – Mar 2021

Data Analyst Intern

- Built an ETL pipeline using Python (Pandas) that replaced 3 hours of daily manual data comparison, increasing efficiency and speeding up data validation.

TECHNICAL SKILLS

Languages & Databases: Python, SQL

Machine Learning & AI: Supervised & Unsupervised Learning, Ensemble Methods (Boosting/Bagging), NLP, Dimensionality Reduction (PCA)

Libraries & Frameworks: Scikit-learn, XGBoost, Pandas, NumPy, FastAPI, NLTK, Matplotlib, Plotly, BeautifulSoup, Streamlit

Tools & BI: Power BI, Git, GitHub, Linux, Command Line

PROJECTS

Crime Communities Predictive Modeling 📊 | Python, Scikit-learn, XGBoost, Pandas, Streamlit

- Developed an end-to-end machine learning pipeline to analyze and predict community crime patterns, applying advanced feature engineering to high-dimensional datasets.
- Trained and evaluated complex ensemble models, including tree-based algorithms, gradient boosting, and model stacking techniques, to optimize predictive accuracy.
- Conducted deep model interpretability analysis using Feature Importance and Partial Dependence Plots (PDPs) to explain the impact of socioeconomic variables on model outputs.
- Deployed the final models as a live, interactive web application using **Streamlit** 📄, enabling real-time predictions and data visualization.

Peloton Evolution: Statistical Mechanics of Cycling 🚴 | Python, Polars, SciPy, Matplotlib

- Engineered a high-performance data pipeline using Polars to parse, clean, and process over 3.01 million observations of UCI Road Race data.
- Applied statistical mechanics and a Monte Carlo simulation (using a two-regime logistic mixture model) to successfully replicate and validate real-world phase transitions in collective motion.
- Modeled non-linear physiological fatigue dynamics and aerodynamic drag reduction, integrating advanced mathematical theory with real-world empirical data.